

ON ASYMMETRIC CONTAMINATION OF GRAPHICAL MODELS

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ABSTRACT. In this paper, we discuss asymmetric graphical models and learn about how we can recover the underlying graph structure via estimating the precision matrix.

1. INTRODUCTION

1.1. **Model.** We present an *Asymmetric Alternative t -distribution* as our proposed graphical model. The model follows a distribution

$$Y = (Y_1, Y_2, \dots, Y_p)$$

with

$$Y_j = \mu_j + \eta_j \nu_j \tau_j + \sqrt{\tau_j} X_j$$

where

$$\tau_j \sim \text{Inv-Gamma}\left(\frac{2}{\nu_j}, \frac{2}{\nu_j}\right), \forall j = 1, 2, \dots, p; \quad \tau_j \perp \tau_k \text{ for } j \neq k$$

$$X = (X_1, X_2, \dots, X_p) \sim \mathcal{N}_p(0, \Psi), \Psi \in \mathbb{R}^{p \times p}$$

and we have the parameter space

$$\Omega = \{(\mu, \eta, \nu) \mid \mu \in \mathbb{R}^p, \eta \in \mathbb{R}^p, \nu \in \mathbb{R}^{+p}\} \cup \{(\mu, \eta, \nu) \mid \mu \in \mathbb{R}^p, \eta = 0, \nu = 0\}$$

where the second set represents the Gaussian case.

Furthermore, we define $\Theta = \Psi^{-1}$ where Θ is the precision matrix.

2. MODEL PROPERTIES

In this section, we highlight some properties on why we care about this model.

2.1. **On identifiability.** The only possible issue of identifiability arises when τ converges to a deterministic constant when $\nu = 0$, which is resolved by the fact that $\nu = 0 \implies \eta = 0$. Therefore, the model is identifiable on the parameter space Ω .

In practice, as ν becomes close to zero, it is difficult to recover the true value of η

2.2. **Mean vector.** In this section, we derive the mean vector of Y

$$\mathbb{E}[Y] = (\mathbb{E}[Y_1], \mathbb{E}[Y_2], \dots, \mathbb{E}[Y_p]).$$

By linearity of expectation, we get

$$\begin{aligned} \mathbb{E}[Y_j] &= \mathbb{E}[\mu_j + \eta_j \nu_j \tau_j + \sqrt{\tau_j} X_j] = \mathbb{E}[\mu_j] + \mathbb{E}[\eta_j \nu_j \tau_j] + \mathbb{E}[\sqrt{\tau_j} X_j] \\ &= \mu_j + \eta_j \nu_j \mathbb{E}[\tau_j] + \mathbb{E}[\sqrt{\tau_j}] \mathbb{E}[X_j] \end{aligned}$$

Using the moment function of inverse gamma distribution and Gaussian distribution, we get

$$\mathbb{E}[Y_j] = \mu_j + \eta_j \nu_j \frac{\frac{2}{\nu_j}}{\frac{2}{\nu_j} - 1} + \left(\frac{2}{\nu_j}\right)^{1/2} \frac{\Gamma(\frac{2}{\nu_j} - \frac{1}{2})}{\Gamma(\frac{2}{\nu_j})}(0) = \boxed{\mu_j + \frac{2\eta_j \nu_j}{2 - \nu_j}}, \quad \nu_j < 2 \quad (2.1)$$

The restriction of $\nu_j < 2$ is due to the fact that $\mathbb{E}[\tau_j]$ is defined for $\frac{2}{\nu_j} > 1$ only.

2.3. Covariance matrix of Y . In this section, we derive the covariance matrix of Y . Using the law of total covariance, we have

$$\text{Cov}(Y) = \mathbb{E}[\text{Cov}(Y \mid \tau)] + \text{Cov}(\mathbb{E}[Y \mid \tau])$$

Notice that, conditioning on τ , we have

$$Y \mid \tau \sim \mathcal{N}_p(\mu + \eta\nu\tau, D(\sqrt{\tau})\Psi D(\sqrt{\tau})). \quad (2.2)$$

Therefore, we get

$$\text{Cov}(Y) = \mathbb{E}[D(\sqrt{\tau})\Psi D(\sqrt{\tau})] + \text{Cov}(\mu + \eta\nu\tau)$$

Now, entry-wise, we have

$$\text{Cov}(Y_i, Y_j) = \psi_{ij}\mathbb{E}[\sqrt{\tau_i\tau_j}] + \eta_i\eta_j\nu_i\nu_j \text{Cov}(\tau_i, \tau_j)$$

Since $\tau_i \perp \tau_j, i \neq j$, we have

$$\text{Cov}(Y_i, Y_j) = \psi_{ij}\mathbb{E}[\sqrt{\tau_i}]\mathbb{E}[\sqrt{\tau_j}], \quad i \neq j.$$

Using the moment function of inverse gamma distribution, we get

$$\text{Cov}(Y_i, Y_j) = \frac{2\psi_{ij}}{\sqrt{\nu_i\nu_j}} \frac{\Gamma\left(\frac{2}{\nu_i} - \frac{1}{2}\right) \Gamma\left(\frac{2}{\nu_j} - \frac{1}{2}\right)}{\Gamma\left(\frac{2}{\nu_i}\right) \Gamma\left(\frac{2}{\nu_j}\right)}, \quad i \neq j$$

For the diagonal entries, we have

$$\text{Var}(Y_k) = \psi_{kk}\mathbb{E}[\tau_k] + \eta_k^2\nu_k^2 \text{Var}(\tau_k) = \frac{2\psi_{kk}}{2 - \nu_k} + \frac{2\eta_k^2\nu_k^3}{(2 - \nu_k)^2(1 - \nu_k)}, \quad 0 < \nu_k < 1.$$

3. LEARNING THE POSTERIOR

3.1. Marginal properties. Before we analyze our graphical model, it is important to analyze the node properties first.

$$Y_j = \mu_j + \nu_j\eta_j\tau_j + \sqrt{\tau_j}X_j$$

This is a multivariate normal mean-variance mixture, though it is more general as we are generating node-specific parameters. We are interested in the probability density function of each random variable at the node. We get

$$f_{Y_j}(y) = \frac{\left(\frac{2}{\nu_j}\right)^{2/\nu_j}}{\Gamma(2/\nu_j)\sqrt{2\pi\Psi_{jj}}} \times \int_0^\infty \tau^{-(2/\nu_j+3/2)} \exp\left[-\frac{(y - \mu_j - \eta_j\nu_j\tau)^2}{2\Psi_{jj}\tau} - \frac{2}{\nu_j\tau}\right] d\tau$$

which simplifies to

$$f_{Y_j}(y) = C_j \exp\left(\frac{\eta_j\nu_j(y - \mu_j)}{\Psi_{jj}}\right) \times \int_0^\infty \tau^{-(2/\nu_j+3/2)} \exp\left[-\frac{A_j}{\tau} - B_j\tau\right] d\tau$$

where

$$A_j = \frac{(y - \mu_j)^2}{2\Psi_{jj}} + \frac{2}{\nu_j}, B_j = \frac{(\eta_j\nu_j)^2}{2\Psi_{jj}}$$

and

$$C_j = \frac{\left(\frac{2}{\nu_j}\right)^{2/\nu_j}}{\Gamma(2/\nu_j)\sqrt{2\pi\Psi_{jj}}}$$

The integral also has a closed form. To derive that, we use a standard Bessel function identity which is as follows

$$\int_0^\infty x^{\lambda-1} e^{-\beta/x - \gamma x} dx = 2 \left(\frac{\beta}{\gamma}\right)^{\lambda/2} K_\lambda(2\sqrt{\beta\gamma})$$

We can apply this identity with $\lambda = -\left(\frac{2}{\nu_j} + \frac{1}{2}\right)$, $\beta = A_j$, and $\gamma = B_j$ to get

$$\int_0^\infty \tau^{-(2/\nu_j+3/2)} \exp\left[-\frac{A_j}{\tau} - B_j\tau\right] d\tau = 2 \left(\frac{A_j}{B_j}\right)^{\lambda/2} K_\lambda(2\sqrt{A_j B_j}),$$

so the marginal density simplifies to

$$f_{Y_j}(y) = 2C_j \left(\frac{A_j}{B_j}\right)^{\lambda/2} \exp\left(\frac{\eta_j \nu_j (y - \mu_j)}{\Psi_{jj}}\right) K_\lambda(2\sqrt{A_j B_j}), \quad \lambda = -\left(\frac{2}{\nu_j} + \frac{1}{2}\right)$$

3.2. Interpreting the precision matrix. When $\vec{Y} = (Y_1, \dots, Y_p)$ is a multivariate Gaussian distribution, the precision matrix Θ has a nice interpretation i.e. $\theta_{jk} = 0 \implies Y_j$ and Y_k are conditionally independent. However, this does not hold for our distribution and actually fails for even a regular Student t-distribution. This is because when $Y \sim N_p(\mu, \Sigma)$, one can factor the joint density into a product of exponentials and conditional independence only holds when the conditional joint density factors. The following equation represents the joint density

$$f(y) \propto \exp\left(-\frac{1}{2}(y - \mu)^\top \Theta (y - \mu)\right) = \exp\left(-\frac{1}{2} \sum_j \theta_{jj}(y_j - \mu_j)^2 - \sum_{j < k} \theta_{jk}(y_j - \mu_j)(y_k - \mu_k)\right)$$

When $\theta_{jk} = 0$, the cross term $\sum_{j < k} \theta_{jk}(y_j - \mu_j)(y_k - \mu_k)$ is 0. Thus, using $\exp(A + B) = \exp(A) \cdot \exp(B)$, we can easily factor this as a product. Moreover, the formula does not have any other cross terms, forcing Y_k and Y_j to be conditionally independent. However, for a t-distribution, we have the following joint density

$$f_\nu(y; \mu, \Psi) \propto \left[1 + \frac{1}{\nu}(y - \mu)^\top \Psi^{-1}(y - \mu)\right]^{-(\nu+p)/2}$$

This can be expanded to

$$f_\nu(y) \propto \left[1 + \frac{1}{\nu} \left(\sum_j \theta_{jj}(y_j - \mu_j)^2 + 2 \sum_{j < k} \theta_{jk}(y_j - \mu_j)(y_k - \mu_k)\right)\right]^{-(\nu+p)/2}$$

When $\theta_{jk} = 0$, although the second term vanishes, one cannot provide a clean factorization for $f_\nu(y)$. Since conditional independence only holds when the conditional joint density factors, we cannot conclude $Y_j \perp Y_k \mid Y_{-j,k}$.

This leaves us with no clean interpretation of the covariance matrix in the context of conditional independence. Thus, we proceed with hypothesis testing. We consider testing whether

two observed variables remain conditionally independent after accounting for all remaining variables. Specifically, for each pair (Y_j, Y_k) we test

$$H_0 : Y_j \perp Y_k \mid Y_{-j,k}$$

against the alternative

$$H_1 : Y_j \not\perp Y_k \mid Y_{-j,k}$$

The algorithm is as follows. First, we estimate the conditional mean functions $\mathbb{E}(Y_i \mid Y_{-i,j})$ and $\mathbb{E}(Y_j \mid Y_{-i,j})$ from our proposed amortized variational inference method. Second, we compute the residuals by subtracting the estimated conditional means from the observed responses. Finally, we adopt the Generalized Covariance Measure which provides us a test statistic to do hypothesis testing on.

$$R_i^{(k)} = Y_i^{(k)} - E[Y_i \mid Y_{-(i,j)}], \text{ and } R_j^{(k)} = Y_j^{(k)} - E[Y_j \mid Y_{-(i,j)}].$$

Under the null hypothesis, we must have $E[R_i R_j] = 0$. Then, we compute the test statistic that is

$$T^{(k)} = \frac{\sqrt{n}\hat{C}}{\hat{\sigma}}$$

where $\hat{C} = \frac{1}{n} \sum_{k=1}^n R_i^{(k)} R_j^{(k)}$ and $\hat{\sigma} = \frac{1}{n} \sum_{k=1}^n (R_i^{(k)} R_j^{(k)})^2 - \hat{C}^2$. GCM theory shows that when $T^{(k)} \rightarrow N_d(0, 1)$ when $n \rightarrow \infty$, then $T^{(k)}$ acts as a normal random variable. Thus, $p = 2(1 - \Phi(|T^{(k)}|))$ where $\Phi(z) = P(Z \leq z)$ where $Z \sim N(0, 1)$

3.3. Conditional Covariance and Conditional Correlation. Recall that

$$Y \mid \tau \sim \mathcal{N}_p(\mu + \eta\nu\tau, D(\sqrt{\tau})\Theta^{-1}D(\sqrt{\tau})).$$

Partition the precision matrix into

$$\Theta = \begin{bmatrix} \Theta_{aa} & \Theta_{ab} \\ \Theta_{ba} & \Theta_{bb} \end{bmatrix}.$$

Denote $K = D(\sqrt{\tau})\Theta^{-1}D(\sqrt{\tau})$. Then we have

$$K = \begin{bmatrix} D_a(\sqrt{\tau})\Theta_{aa}D_a(\sqrt{\tau}) & D_a(\sqrt{\tau})\Theta_{ab}D_b(\sqrt{\tau}) \\ D_b(\sqrt{\tau})\Theta_{ba}D_a(\sqrt{\tau}) & D_b(\sqrt{\tau})\Theta_{bb}D_b(\sqrt{\tau}) \end{bmatrix} = \begin{bmatrix} K_{aa} & K_{ab} \\ K_{ba} & K_{bb} \end{bmatrix}$$

Using standard Gaussian results (Appendix 8.1), we have

$$Y_a \mid Y_b, \tau \sim \mathcal{N}_2(m_{a|b,\tau}, D_a(\sqrt{\tau})\Theta_{aa}^{-1}D_a(\sqrt{\tau}))$$

where

$$\begin{aligned} m_{a|b,\tau} &= \mu_a + \eta_a\nu_a\tau_a - K_{aa}^{-1}K_{ab}(y_b - \mu_b - \eta_b\nu_b\tau_b) \\ &= \mu_a + \eta_a\nu_a\tau_a - D_a(\sqrt{\tau})\Theta_{aa}^{-1}\Theta_{ab}D_b(1/\sqrt{\tau})(y_b - \mu_b - \eta_b\nu_b\tau_b) \end{aligned}$$

Now, we can conclude that

$$\text{Cov}(Y_i, Y_j \mid Y_{-ij}, \tau) = \sqrt{\tau_i\tau_j}(\Theta_{aa}^{-1})_{ij}$$

Since

$$\Theta_{aa}^{-1} = \begin{bmatrix} \theta_{ii} & \theta_{ij} \\ \theta_{ij} & \theta_{jj} \end{bmatrix}^{-1} = \frac{1}{\theta_{ii}\theta_{jj} - \theta_{ij}^2} \begin{bmatrix} \theta_{jj} & -\theta_{ij} \\ -\theta_{ij} & \theta_{ii} \end{bmatrix} = \frac{1}{\Delta_{ij}} \begin{bmatrix} \theta_{jj} & -\theta_{ij} \\ -\theta_{ij} & \theta_{ii} \end{bmatrix}$$

we have

$$\text{Cov}(Y_i, Y_j \mid Y_{-ij}, \tau) = -\frac{\theta_{ij}\sqrt{\tau_i\tau_j}}{\Delta_{ij}}$$

Using law of total covariance

$$\text{Cov}(Y_i, Y_j \mid Y_{-ij}) = \text{Cov}(\mathbb{E}[Y_i \mid Y_b, \tau], \mathbb{E}[Y_j \mid Y_b, \tau]) + \mathbb{E}[\text{Cov}(Y_i, Y_j \mid Y_b, \tau)],$$

we can integrate out the latent variable τ

$$\text{Cov}(Y_i, Y_j \mid Y_{-ij}) = \text{Cov}(m_{i|b,\tau}, m_{j|b,\tau} \mid Y_b) - \frac{\theta_{ij}}{\Delta_{ij}} \mathbb{E}[\sqrt{\tau_i \tau_j}]$$

Notice that when $\theta_{ij} = 0$, the second term vanish, so we study the first term.

Computing $\text{Cov}(m_{i|b,\tau}, m_{j|b,\tau})$:

Expanding $m_{a|b,\tau}$ gives

$$m_{a|b,\tau} = \mu_a + \eta_a \nu_a \tau_a - \frac{1}{\Delta_{ij}} \begin{bmatrix} \sqrt{\tau_i} & 0 \\ 0 & \sqrt{\tau_j} \end{bmatrix} \begin{bmatrix} \theta_{jj} & -\theta_{ij} \\ -\theta_{ij} & \theta_{ii} \end{bmatrix} \begin{bmatrix} \theta_{i,b}^T \\ \theta_{j,b}^T \end{bmatrix} D_b(1/\sqrt{\tau})(y_b - \mu_b - \eta_b \nu_b \tau_b).$$

Doing matrix multiplications give

$$m_{a|b,\tau} = \mu_a + \eta_a \nu_a \tau_a - \frac{1}{\Delta_{ij}} \begin{bmatrix} \left(\sqrt{\tau_i} \theta_{jj} \widetilde{\theta}_{i,b}^T - \sqrt{\tau_i} \theta_{ij} \widetilde{\theta}_{j,b}^T \right) (y_b - \mu_b - \eta_b \nu_b \tau_b) \\ \left(\sqrt{\tau_j} \theta_{ii} \widetilde{\theta}_{j,b}^T - \sqrt{\tau_j} \theta_{ij} \widetilde{\theta}_{i,b}^T \right) (y_b - \mu_b - \eta_b \nu_b \tau_b) \end{bmatrix},$$

where

$$\begin{aligned} \widetilde{\theta}_{i,b}^T &= \theta_{i,b}^T D_b(1/\sqrt{\tau}) \\ \widetilde{\theta}_{j,b}^T &= \theta_{j,b}^T D_b(1/\sqrt{\tau}). \end{aligned}$$

Thus, we get that

$$\begin{aligned} m_{i|b,\tau} &= \mu_i + \eta_i \nu_i \tau_i - \frac{\sqrt{\tau_i}}{\Delta_{ij}} \left(\theta_{jj} \widetilde{\theta}_{i,b}^T - \theta_{ij} \widetilde{\theta}_{j,b}^T \right) (y_b - \mu_b - \eta_b \nu_b \tau_b) \\ m_{j|b,\tau} &= \mu_j + \eta_j \nu_j \tau_j - \frac{\sqrt{\tau_j}}{\Delta_{ij}} \left(\theta_{ii} \widetilde{\theta}_{j,b}^T - \theta_{ij} \widetilde{\theta}_{i,b}^T \right) (y_b - \mu_b - \eta_b \nu_b \tau_b) \end{aligned}$$

Assuming $\theta_{ij} = 0$, we have

$$\begin{aligned} m_{i|b,\tau} &= \mu_i + \eta_i \nu_i \tau_i - \frac{\sqrt{\tau_i}}{\theta_{ii}} \left(\widetilde{\theta}_{i,b}^T \right) (y_b - \mu_b - \eta_b \nu_b \tau_b) \\ m_{j|b,\tau} &= \mu_j + \eta_j \nu_j \tau_j - \frac{\sqrt{\tau_j}}{\theta_{jj}} \left(\widetilde{\theta}_{j,b}^T \right) (y_b - \mu_b - \eta_b \nu_b \tau_b). \end{aligned}$$

For simplicity of covariance computation, define

$$h_p(\tau_p) = \frac{y_p - \mu_p}{\sqrt{\tau_p}} - \eta_p \nu_p \sqrt{\tau_p}.$$

The conditional means can then be written as

$$\begin{aligned} m_{i|b,\tau} &= \mu_i + \eta_i \nu_i \tau_i - \frac{\sqrt{\tau_i}}{\theta_{ii}} \sum_{p \in b} \theta_{ip} h_p(\tau_p), \\ m_{j|b,\tau} &= \mu_j + \eta_j \nu_j \tau_j - \frac{\sqrt{\tau_j}}{\theta_{jj}} \sum_{q \in b} \theta_{jq} h_q(\tau_q). \end{aligned}$$

Now, using the bilinearity of covariance, we get

$$\text{Cov}(m_{i|b,\tau}, m_{j|b,\tau} \mid Y_b) = \eta_i \nu_i \eta_j \nu_j \text{Cov}(\tau_i, \tau_j \mid Y_b) \quad (\text{A})$$

$$- \frac{\eta_i \nu_i}{\theta_{jj}} \sum_{q \in b} \theta_{jq} \text{Cov}(\tau_i, \sqrt{\tau_j} h_q(\tau_q) \mid Y_b) \quad (\text{B})$$

$$- \frac{\eta_j \nu_j}{\theta_{ii}} \sum_{p \in b} \theta_{ip} \text{Cov}(\sqrt{\tau_i} h_p(\tau_p), \tau_j \mid Y_b) \quad (\text{C})$$

$$+ \frac{1}{\theta_{ii} \theta_{jj}} \sum_{p \in b} \sum_{q \in b} \theta_{ip} \theta_{jq} \text{Cov}(\sqrt{\tau_i} h_p(\tau_p), \sqrt{\tau_j} h_q(\tau_q) \mid Y_b). \quad (\text{D})$$

By independence between coordinates of τ , it's easy to see that (A),(B),(C) are all zero. Thus, we are left with

$$\begin{aligned} \text{Cov}(m_{i|b,\tau}, m_{j|b,\tau} \mid Y_b) &= \frac{1}{\theta_{ii} \theta_{jj}} \sum_{p \in b} \sum_{q \in b} \theta_{ip} \theta_{jq} \text{Cov}(\sqrt{\tau_i} h_p(\tau_p), \sqrt{\tau_j} h_q(\tau_q) \mid Y_b) \\ &= \frac{\mathbb{E}[\sqrt{\tau_i}] \mathbb{E}[\sqrt{\tau_j}]}{\theta_{ii} \theta_{jj}} \sum_{p \in b} \sum_{q \in b} \theta_{ip} \theta_{jq} \text{Cov}(h_p(\tau_p), h_q(\tau_q) \mid Y_b). \end{aligned}$$

4. INFERENCE

4.1. MCMC.

4.2. Variational EM. In the previous section, we derived the complete-data log-likelihood $\ell_{\text{hid}}(\mu, \eta, \nu, \Theta \mid Y, \tau)$, treating the latent scale vectors $\tau^{(i)} = (\tau_1^{(i)}, \dots, \tau_p^{(i)})$ as observed. In a standard EM algorithm, the E-step would require computing

$$Q(\theta \mid \theta^{(t)}) = \mathbb{E}_{\tau \mid Y, \theta^{(t)}}[\ell_{\text{hid}}(\theta \mid Y, \tau)],$$

which in turn requires the exact posterior $p(\tau^{(i)} \mid Y^{(i)}, \theta^{(t)})$. When Θ is diagonal, this posterior factors coordinatewise into a tractable Generalized Inverse Gaussian (GIG) distribution for each $\tau_j^{(i)}$. However, for a general positive-definite Θ , the cross terms $\theta_{jk} z_j^{(i)} z_k^{(i)}$ ($j \neq k$) couple the coordinates of $\tau^{(i)}$, and the resulting joint posterior no longer belongs to a tractable family. To retain a closed-form E-step, we instead use variational approximation of this posterior.

4.2.1. Variational approximation of the conditional density. When we condition on $\tau^{(i)}$, we get

$$Y^{(i)} \mid \tau^{(i)} \sim N_p\left(\mu + \eta \nu \tau^{(i)}, D(\sqrt{\tau^{(i)}}) \Theta^{-1} D(\sqrt{\tau^{(i)}})\right)$$

and we approximate this distribution by the closest member, using KL-divergence, of the family of multivariate normals with diagonal covariance. This allows us to retain a factorable density. This approximated distribution is

$$Y^{(i)} \mid \tau^{(i)} \sim \mathcal{N}_p\left(\mu + \eta \nu \tau^{(i)}, D(\sqrt{\tau^{(i)}}) \bar{\Theta}^{-1} D(\sqrt{\tau^{(i)}})\right).$$

where $\bar{\Theta} = \text{diag}(\bar{\theta}_{11}, \dots, \bar{\theta}_{pp})$ is exactly the diagonal of Θ . This gives us the approximating conditional density

$$q^*(Y^{(i)} | \tau^{(i)}) = \prod_{j=1}^p q^*(Y_j^{(i)} | \tau_j^{(i)}), \quad Y_j^{(i)} | \tau_j^{(i)} \sim N\left(\mu_j + \eta_j \nu_j \tau_j^{(i)}, \frac{\tau_j^{(i)}}{\theta_{jj}}\right).$$

Because the latent variables are themselves independent across coordinates ($f(\tau^{(i)}) = \prod_j f(\tau_j^{(i)})$), the approximate posterior obtained via Bayes' rule,

$$q^*(\tau^{(i)} | Y^{(i)}) \propto q^*(Y^{(i)} | \tau^{(i)}) f(\tau^{(i)}),$$

also factors across j . Matching the resulting kernel in $\tau_j^{(i)}$ to the GIG density gives

$$\tau_j^{(i)} | Y_j^{(i)} \sim \text{GIG}(\lambda_j, \chi_{ij}, \psi_j), \quad \lambda_j = -\frac{2}{\nu_j} - \frac{1}{2}, \quad \chi_{ij} = \frac{4}{\nu_j} + \bar{\theta}_{jj}(Y_j^{(i)} - \mu_j)^2, \quad \psi_j = \bar{\theta}_{jj} \eta_j^2 \nu_j^2.$$

This approximate posterior is what we use in place of the (intractable, for general Θ) exact posterior throughout the E-step. We direct the reader to the appendix to see a full derivation of the conditional density.

4.2.2. Expectation of the posterior. Writing $z_j^{(i)} = (\tau_j^{(i)})^{-1/2}(Y_j^{(i)} - \mu_j) - (\tau_j^{(i)})^{1/2}\gamma_j$, with $\gamma := \eta\nu$, the complete-data log-likelihood depends on $\tau^{(i)}$ only through the terms $\tau_j^{(i)}$, $1/\tau_j^{(i)}$, $\log \tau_j^{(i)}$ (from the Inv-Gamma prior term) and through $z_j^{(i)} z_k^{(i)}$ (from $\text{Tr}(S_\tau \Theta)$). Since $z_j^{(i)}$ depends on $\tau_j^{(i)}$ alone, and the approximate posterior is coordinatewise independent, all of these reduce to functions of the moments

$$M_{ij}(r) := \mathbb{E}\left[(\tau_j^{(i)})^r | Y_j^{(i)}\right] = \left(\frac{\chi_{ij}}{\psi_j}\right)^{r/2} \frac{K_{\lambda_j+r}(\sqrt{\chi_{ij}\psi_j})}{K_{\lambda_j}(\sqrt{\chi_{ij}\psi_j})},$$

a standard property of the GIG family, where K_λ is the modified Bessel function of the second kind. In particular $\mathbb{E}[\tau_j^{(i)}] = M_{ij}(1)$, $\mathbb{E}[1/\tau_j^{(i)}] = M_{ij}(-1)$, $\mathbb{E}[\sqrt{\tau_j^{(i)}}] = M_{ij}(1/2)$, and $\mathbb{E}[1/\sqrt{\tau_j^{(i)}}] = M_{ij}(-1/2)$, giving closed-form expressions for $\mathbb{E}[z_j^{(i)} z_k^{(i)} | Y^{(i)}]$.

The logarithmic moment $\mathbb{E}[\log \tau_j^{(i)} | Y_j^{(i)}]$ has no elementary closed form, since it involves $\partial_\lambda \log K_\lambda$. We approximate it by central finite differences in λ_j :

$$L_{ij}(h) := \frac{1}{2} \log\left(\frac{\chi_{ij}}{\psi_j}\right) + \frac{\log K_{\lambda_j+h}(\sqrt{\chi_{ij}\psi_j}) - \log K_{\lambda_j-h}(\sqrt{\chi_{ij}\psi_j})}{2h}, \quad \mathbb{E}[\log \tau_j^{(i)} | Y_j^{(i)}] \approx L_{ij}(h).$$

Substituting $M_{ij}(\cdot)$ and $L_{ij}(h)$ into ℓ_{hid} in place of the corresponding latent quantities gives the variational surrogate objective $Q(\theta | \theta^{(t)})$, which is now a closed-form function of $\theta = (\mu, \eta, \nu, \Theta)$ alone.

4.2.3. Maximization. We maximize $Q(\theta | \theta^{(t)})$ using an ECM-type blockwise update. In this approach, when optimizing μ, η, ν , we retain our approximation of Θ by the diagonal matrix $\bar{\Theta}$.

Optimizing μ and γ

Under the above approximation, the trace term then reduces to

$$\text{Tr}(S_\tau \Theta) = \frac{1}{n} \sum_{i=1}^n \sum_{j=1}^p \theta_{jj} (z_j^{(i)})^2 = \frac{1}{n} \sum_{i=1}^n \sum_{j=1}^p \theta_{jj} (z_j^{(i)})^2.$$

Then we have

$$\mathbb{E}[\text{Tr}(S_\tau \Theta) \mid Y^{(1)}, Y^{(2)}, \dots, Y^{(n)}] = \text{Tr}(\hat{S}_\tau \Theta) = \frac{1}{n} \sum_{j=1}^p \theta_{jj} \sum_{i=1}^n \mathbb{E}[(z_j^{(i)})^2 \mid Y_j^{(i)}],$$

which we aim to minimize. Since

$$Q(\mu_j, \gamma_j) := \sum_{i=1}^n \mathbb{E}[(z_j^{(i)})^2 \mid Y_j^{(i)}] = \sum_{i=1}^n M_{ij}(-1)(Y_j^{(i)} - \mu_j)^2 + M_{ij}(1)\gamma_j^2 - 2(Y_j^{(i)} - \mu_j)\gamma_j$$

is a convex quadratic function of μ_j, γ_j , we optimize it by finding the zeros of the partial derivatives with respect to μ_j, γ_j

$$\begin{aligned} \frac{\partial Q}{\partial \mu_j} &= \sum_{i=1}^n 2 \left[\gamma_j - M_{ij}(-1) (Y_j^{(i)} - \mu_j) \right] = 0 \\ \frac{\partial Q}{\partial \gamma_j} &= \sum_{i=1}^n 2 \left[M_{ij}(1)\gamma_j - (Y_j^{(i)} - \mu_j) \right] = 0 \end{aligned}$$

which gives

$$\begin{pmatrix} \sum_{i=1}^n M_{ij}(-1) & n \\ n & \sum_{i=1}^n M_{ij}(1) \end{pmatrix} \begin{pmatrix} \mu_j \\ \gamma_j \end{pmatrix} = \begin{pmatrix} \sum_{i=1}^n M_{ij}(-1)Y_j^{(i)} \\ \sum_{i=1}^n Y_j^{(i)} \end{pmatrix}.$$

Let

$$A_j = \sum_{i=1}^n M_{ij}(-1), \quad B_j = \sum_{i=1}^n M_{ij}(1), \quad R_j = \sum_{i=1}^n M_{ij}(-1)Y_j^{(i)}, \quad T_j = \sum_{i=1}^n Y_j^{(i)}.$$

We then get

$$\begin{pmatrix} A_j & n \\ n & B_j \end{pmatrix} \begin{pmatrix} \mu_j \\ \gamma_j \end{pmatrix} = \begin{pmatrix} R_j \\ T_j \end{pmatrix},$$

solving which gives

$$\boxed{\mu_j^{(t+1)} = \frac{B_j R_j - n T_j}{A_j B_j - n^2}, \quad \gamma_j^{(t+1)} = \frac{A_j T_j - n R_j}{A_j B_j - n^2}.$$

Optimizing ν

To optimize ν , we maximize

$$\sum_{j=1}^p Q(\nu_j),$$

where

$$Q(\nu_j) := \mathbb{E} \left[\sum_{i=1}^n \left\{ \frac{2}{\nu_j} \log \left(\frac{2}{\nu_j} \right) - \log \Gamma \left(\frac{2}{\nu_j} \right) - \left(\frac{2}{\nu_j} \right) \log \tau_j^{(i)} - \frac{2}{\nu_j \tau_j^{(i)}} \right\} \right]$$

which can be simplified into

$$Q(\nu_j) = n \left[\frac{2}{\nu_j} \log \left(\frac{2}{\nu_j} \right) - \log \Gamma \left(\frac{2}{\nu_j} \right) \right] - \frac{2}{\nu_j} \sum_{i=1}^n (L_{ij}(h) + M_{ij}(-1)).$$

Differentiating with respect to ν_j gives

$$\frac{\partial Q}{\partial \nu_j} = -\frac{2}{\nu_j^2} \left[n \left(\log \left(\frac{2}{\nu_j} \right) + 1 - \psi \left(\frac{2}{\nu_j} \right) \right) - \sum_{i=1}^n (L_{ij}(h) + M_{ij}(-1)) \right],$$

where ψ is the digamma function. Thus, $\nu_j^{(t+1)}$ is updated by numerically solving

$$n \left(\log \left(\frac{2}{\nu_j} \right) + 1 - \psi \left(\frac{2}{\nu_j} \right) \right) - \sum_{i=1}^n (L_{ij}(h) + M_{ij}(-1)) = 0.$$

Recovering η

With $\gamma^{(t+1)}, \nu^{(t+1)}$ updated, we can recover $\eta^{(t+1)}$ by

$$\eta^{(t+1)} = \frac{\gamma^{(t+1)}}{\nu^{(t+1)}}$$

Optimizing Θ

After updating $\mu^{(t+1)}, \eta^{(t+1)}, \nu^{(t+1)}$, we solve the optimization problem

$$\arg \max_{\Theta \succ 0} \left\{ \log \det \Theta - \text{Tr}(\hat{S}_\tau(\mu^{(t+1)}, \eta^{(t+1)}, \nu^{(t+1)})\Theta) - \rho|\Theta| \right\}$$

which is the standard *lasso* objective. Each block update increases (or does not decrease) Q , so the overall scheme is a valid generalized EM algorithm, converging to a stationary point of the variational surrogate objective.

5. SIMULATION

For our simulation, we estimated the covariance matrix with MCMC and Variational EM. Moreover, we report the estimated covariance matrix computed at the edge cases of each parameter.

6. APPLICATIONS

We test our model on

7. CONCLUSION

8. APPENDIX

8.1. Conditional Marginal Gaussian Distribution.

Let

$$Y \sim \mathcal{N}(\mu, \Theta^{-1})$$

Then we have the pdf

$$p(y) \propto \exp \left\{ -\frac{1}{2}(y - \mu)^T \Theta (y - \mu) \right\}$$

Let $a = \{i, j\}, b = V \setminus \{i, j\}$, we then get:

$$p(y_a \mid y_b) \propto \exp \left\{ -\frac{1}{2} (y - \mu)^T \Theta (y - \mu) \right\}$$

Notice that we can expand this and keep only the part that involves y_a

$$\begin{aligned} (y - \mu)^T \Theta (y - \mu) &= [(y_a - \mu_a)^T \quad (y_b - \mu_b)^T] \begin{bmatrix} \Theta_{aa} & \Theta_{ab} \\ \Theta_{ba} & \Theta_{bb} \end{bmatrix} \begin{bmatrix} y_a - \mu_a \\ y_b - \mu_b \end{bmatrix} \\ &= (y_a - \mu_a)^T \Theta_{aa} (y_a - \mu_a) + 2(y_a - \mu_a)^T \Theta_{ab} (y_b - \mu_b) + (y_b - \mu_b)^T \Theta_{bb} (y_b - \mu_b) \\ &\propto (y_a - \mu_a)^T \Theta_{aa} (y_a - \mu_a) + 2(y_a - \mu_a)^T \Theta_{ab} (y_b - \mu_b) \end{aligned}$$

We can then complete the square using $x = y_a - \mu_a, c = \Theta_{ab}(y_b - \mu_b)$

$$(y - \mu)^T \Theta (y - \mu) \propto (x + \Theta_{aa}^{-1} c)^T \Theta_{aa} (x + \Theta_{aa}^{-1} c)$$

Therefore, we get

$$p(y_a \mid y_b, \tau) = \mathcal{N}_2(\mu_a - \Theta_{aa}^{-1} c, \Theta_{aa}^{-1}) = \mathcal{N}_2(\mu_a - \Theta_{aa}^{-1} \Theta_{ab} (y_b - \mu_b), \Theta_{aa}^{-1})$$